Type MS · Rail Fence cipher

Answer 1 of 5 · Question 3 part c.i · 4 marks

2020 NOV · Paper 02 · extract 1 of 6 · find part c.i in this mark scheme extract

3 (a)	A1	Award one mark for any of: • Conversion of plain text into cipher (1) • Converting information/data into a code (1) • Converting information/data into an unreadable format (1)	Accept alternative wording	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 1 of 5 · Question 3 part c.i · 4 marks

2020 NOV · Paper 02 · extract 2 of 6 · find part c.i in this mark scheme extract

3 (b)	B1	Award one mark for any of: • To ensure that the data can only be read by an authorised person / can't be read by an unauthorised person. (1)	Accept alternative similar wording. Do not accept hacking.	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 1 of 5 · Question 3 part c.i · 4 marks

2020 NOV · Paper 02 · extract 3 of 6 · find part c.i in this mark scheme extract

3 (c) (i)	C1 C2 C3 C4	Award 4 marks for a correct response: • CAL|OTTA|MUIN|PO • PO|MUIN|OTTA|CAL Award 1 mark each up to a maximum of 3 for: • 4 lines of text (1) • Zigzag arrangement of letters (1) • Letters from each line reproduced / consistent use of the key 4: GSQTYXEXMSREP/ YKILQPWPEKJWH (1)		(4)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 1 of 5 · Question 3 part c.i · 4 marks

2020 NOV · Paper 02 · extract 4 of 6 · find part c.i in this mark scheme extract

3 (c) (ii)	C5 C6	Award 2 marks for a linked explanation such as: • Limited number of usable keys (1) to allow for sufficient movement of characters / so can easily be decoded by trial		(2)

5 of 16

Type MS · Rail Fence cipher

Answer 1 of 5 · Question 3 part c.i · 4 marks

2020 NOV · Paper 02 · extract 5 of 6 · find part c.i in this mark scheme extract

		and error / easy to use brute force to crack (1)		

6 of 16

Type MS · Rail Fence cipher

Answer 1 of 5 · Question 3 part c.i · 4 marks

2020 NOV · Paper 02 · extract 6 of 6 · find part c.i in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 2 of 5 · Question 3 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 1 of 6 · find part c.ii in this mark scheme extract

3 (a)	A1	Award one mark for any of: • Conversion of plain text into cipher (1) • Converting information/data into a code (1) • Converting information/data into an unreadable format (1)	Accept alternative wording	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 2 of 5 · Question 3 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 2 of 6 · find part c.ii in this mark scheme extract

3 (b)	B1	Award one mark for any of: • To ensure that the data can only be read by an authorised person / can't be read by an unauthorised person. (1)	Accept alternative similar wording. Do not accept hacking.	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 2 of 5 · Question 3 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 3 of 6 · find part c.ii in this mark scheme extract

3 (c) (i)	C1 C2 C3 C4	Award 4 marks for a correct response: • CAL|OTTA|MUIN|PO • PO|MUIN|OTTA|CAL Award 1 mark each up to a maximum of 3 for: • 4 lines of text (1) • Zigzag arrangement of letters (1) • Letters from each line reproduced / consistent use of the key 4: GSQTYXEXMSREP/ YKILQPWPEKJWH (1)		(4)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 2 of 5 · Question 3 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 4 of 6 · find part c.ii in this mark scheme extract

3 (c) (ii)	C5 C6	Award 2 marks for a linked explanation such as: • Limited number of usable keys (1) to allow for sufficient movement of characters / so can easily be decoded by trial		(2)

5 of 16

Type MS · Rail Fence cipher

Answer 2 of 5 · Question 3 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 5 of 6 · find part c.ii in this mark scheme extract

		and error / easy to use brute force to crack (1)		

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Type MS · Rail Fence cipher

Answer 2 of 5 · Question 3 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 6 of 6 · find part c.ii in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 3 of 5 · Question 4 part c.i · 1 mark

2022 JUN · Paper 02 · extract 1 of 6 · find part c.i in this mark scheme extract

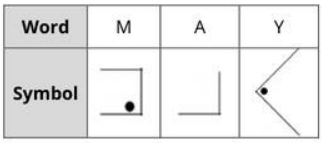
4 (a)	A1	**The only correct answer is B** A is not correct because this is decryption C is not correct because this is not an encryption D is not correct because this is not an encryption		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 3 of 5 · Question 4 part c.i · 1 mark

2022 JUN · Paper 02 · extract 2 of 6 · find part c.i in this mark scheme extract

4 (b)	B1 B2 B3	**Award 1 mark for each correct symbol**  		(3)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 3 of 5 · Question 4 part c.i · 1 mark

2022 JUN · Paper 02 · extract 3 of 6 · find part c.i in this mark scheme extract



Type MS · Rail Fence cipher

Answer 3 of 5 · Question 4 part c.i · 1 mark

2022 JUN · Paper 02 · extract 4 of 6 · find part c.i in this mark scheme extract



Type MS · Rail Fence cipher

Answer 3 of 5 · Question 4 part c.i · 1 mark

2022 JUN · Paper 02 · extract 5 of 6 · find part c.i in this mark scheme extract

4 (c)(iii)	C3	Rail (Fence cipher)		(1)

Type MS · Rail Fence cipher

Answer 3 of 5 · Question 4 part c.i · 1 mark

2022 JUN · Paper 02 · extract 6 of 6 · find part c.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 4 of 5 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 02 · extract 1 of 6 · find part c.ii in this mark scheme extract

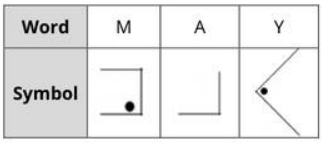
4 (a)	A1	**The only correct answer is B** A is not correct because this is decryption C is not correct because this is not an encryption D is not correct because this is not an encryption		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 4 of 5 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 02 · extract 2 of 6 · find part c.ii in this mark scheme extract

4 (b)	B1 B2 B3	**Award 1 mark for each correct symbol**  		(3)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 4 of 5 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 02 · extract 3 of 6 · find part c.ii in this mark scheme extract

4 (c)(ii)	C1	5		(1)

Type MS · Rail Fence cipher

Answer 4 of 5 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 02 · extract 4 of 6 · find part c.ii in this mark scheme extract



Type MS · Rail Fence cipher

Answer 4 of 5 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 02 · extract 5 of 6 · find part c.ii in this mark scheme extract

4 (c)(iii)	C3	Rail (Fence cipher)		(1)

Type MS · Rail Fence cipher

Answer 4 of 5 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 02 · extract 6 of 6 · find part c.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 5 of 5 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 02 · extract 1 of 6 · find part c.iii in this mark scheme extract

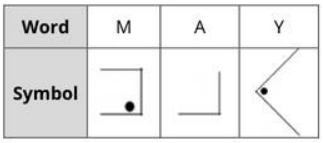
4 (a)	A1	**The only correct answer is B** A is not correct because this is decryption C is not correct because this is not an encryption D is not correct because this is not an encryption		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 5 of 5 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 02 · extract 2 of 6 · find part c.iii in this mark scheme extract

4 (b)	B1 B2 B3	**Award 1 mark for each correct symbol**  		(3)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Rail Fence cipher

Answer 5 of 5 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 02 · extract 3 of 6 · find part c.iii in this mark scheme extract

4 (c)(i)	C1	5		(1)

Type MS · Rail Fence cipher

Answer 5 of 5 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 02 · extract 4 of 6 · find part c.iii in this mark scheme extract



Type MS · Rail Fence cipher

Answer 5 of 5 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 02 · extract 5 of 6 · find part c.iii in this mark scheme extract

4 (c)(iii)	C3	Rail (Fence cipher)		(1)

Type MS · Rail Fence cipher

Answer 5 of 5 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 02 · extract 6 of 6 · find part c.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark